

---

**Algorithm 1** Slider recovery decision – from element inventory to recovered set

---

**Require:** Slider  $s \in [0, 1]$ ; element table  $\{c_i, f_i, V_i, H_i, \eta_i, \pi_i\}$ ; Sherwood constants  $K, b$ ; dry mass  $M_{\text{dry}}$

**Ensure:** Recovered set  $\mathcal{R}(s)$ , recovered masses  $\{m_i\}$ , slag-bound masses  $\{r_i\}$

```
1:  $\mathcal{R} \leftarrow \emptyset$ 
2: for each element  $i$  do
3:    $c_i^{\text{eff}} \leftarrow \min(1, f_i c_i)$ ;    $C_i \leftarrow K (c_i^{\text{eff}})^{-b}$  ▷ Sherwood cost, Eq. 2
4:   if  $\pi_i = \text{IMMOBILIZE}$  then
5:      $r_i \leftarrow c_i M_{\text{dry}}$  ▷ policy exclusion: stays in glass
6:   else if  $\pi_i = \text{ALWAYS}$  or  $V_i + s H_i > C_i$  then
7:      $\mathcal{R} \leftarrow \mathcal{R} \cup \{i\}$ ;    $m_i \leftarrow \eta_i c_i M_{\text{dry}}$ ;    $r_i \leftarrow (1 - \eta_i) c_i M_{\text{dry}}$ 
8:   else
9:      $r_i \leftarrow c_i M_{\text{dry}}$  ▷ below threshold: vitrify
10: return  $\mathcal{R}, \{m_i\}, \{r_i\}$  ▷ feed to mass/energy balance (Sec. 6)
```

---